

CHRONIC KIDNEY DISEASE (CKD) CLINICAL REPORT

Patient Name:	Jane Smith	Patient Hospital ID:	PT-9838
Age / Gender:	45 yrs / Female	Diagnosis Date:	22 May 2026 19:40
Contact Info:	jane@example.com	Attending Physician:	Dr. Administrator

1. Primary Renal Markers & KDIGO Risk Staging

eGFR Value	GFR Stage	Urine ACR	Albuminuria Stage	KDIGO Risk Category
56.89 mL/min/1.73 m²	G3a	150.0 mg/g	A2	HIGH RISK

2. Decision Tree Machine Learning Classifier

Inference Engine Target	Diagnostic Classification	Inference Confidence Score
UCI CKD Trained Classifier Model	CKD DETECTED	100.0%

3. Expanded Urinalysis, Blood Chemistry & Vitals Grid

Parameter Name	Patient Metric	Parameter Name	Patient Metric
Blood Pressure:	120 mmHg	Blood Urea (BUN):	40.0 mg/dL
Hypertension:	no	Serum Creatinine:	1.2 mg/dL
Diabetes Mellitus:	no	Serum Sodium:	138.0 mEq/L
Urine Specific Gravity:	1.02	Serum Potassium:	4.2 mEq/L
Albumin Level (Dipstick):	1 +	Hemoglobin Count:	14.5 g/dL
Sugar Level (Dipstick):	0	White Blood Cell (WBC):	8000 cells/mcL
Red Blood Cells (RBC):	normal	Estimated eGFR:	56.89 mL/min

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4. Gemini AI Clinical Assistant Insights & Protocol

Medical Insights & Critical Findings:

Based on the Decision Tree model trained on the UCI CKD dataset, the patient is classified as 'CKD Detected' with a confidence of 100%. Serum Creatinine (1.2 mg/dL) is elevated above normal gender thresholds, pointing towards reduced glomerular filtration.

Pathophysiological Risk Analysis:

Risk analysis suggests active kidney damage or impaired filtration function. The presence of risk factors needs urgent monitoring to prevent progressive loss of nephrons.

Suggested Precautions	Lifestyle Advice	Follow-Up Protocol
- Review all prescriptions; avoid nephrotoxic drugs, especially NSAIDs (e.g., ibuprofen, naproxen). - Ensure adequate hydration, matching fluid intake with output levels. - Consult a nephrologist if GFR drops below 30 mL/min/1.73 m².	- Maintain low-sodium dietary habits to protect baseline vascular pressure. - Engage in moderate physical activity for 30 minutes, 5 days a week. - Cease smoking immediately, as nicotine accelerates renal microvascular deterioration. - Limit high-protein diets; target moderate intake (0.8 g/kg body weight) to reduce glomerular load.	- Re-check Serum Creatinine and Urine ACR in 3 months to monitor GFR trajectory. - Schedule an ultrasound scan of the kidneys to check for structural changes or scarring. - Maintain an active log of clinical vitals for review at the next outpatient clinic visit.

5. Primary Clinical Guideline & Recommendations

High risk of CKD progression. Refer to nephrology if not already under specialist care. Intensify blood pressure and glycaemic control. Monitor eGFR and ACR every 3–4 months. Evaluate for complications (anaemia, bone disease, acidosis).

Medical Disclaimer: This clinical summary is an educational decision-support tool. It has been generated by combining historical patient classification metrics, clinical KDIGO rules, and artificial intelligence models. Vitals must be checked and confirmed by a certified medical professional before clinical operations are enacted.

Physician Signature:
